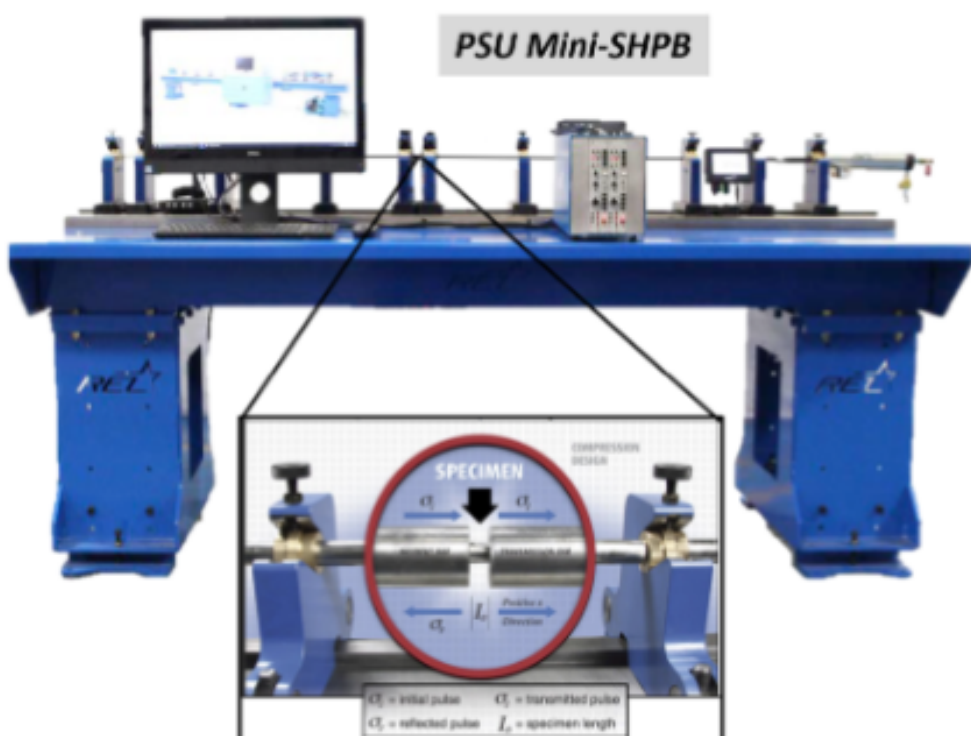


Miniature Kolsky Bar Apparatus (Ø3.17 mm)

EXPERIMENTAL APPARATUS



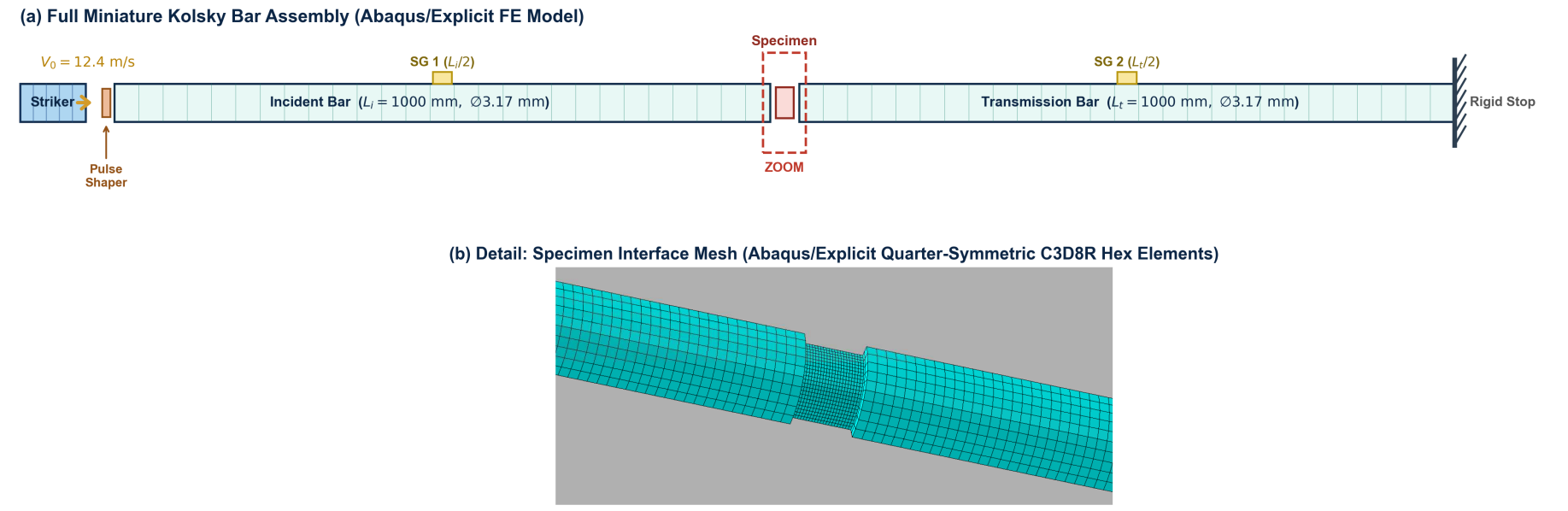
HARDWARE ARCHITECTURE

Bar Material: C350 Maraging Steel
Bar Diameter: Ø3.17 mm (1/8" OD)
Bar Lengths: Incident 1.0 m, Trans. 1.0 m
Wave Speed: $C_0 \approx 4960$ m/s
Striker Range: $C_0 \approx 200$ mm, $V_0 \leq 30$ m/s
Pulse Duration: $T_p \approx 78$ μ s
Gauges: Semiconductor (10 MHz DAQ)
Max Strain Rate: $\dot{\epsilon}_{max} \sim 1.2 \times 10^4$ s⁻¹

- Ultra-Small Diameter Architecture:** Precision Ø3.17 mm C350 maraging steel bar system specifically engineered for sub-scale testing of advanced composites, ductile metals (OFHC copper), and transparent polymers (PMMA).
- Extreme Dynamic Strain Rate Regime:** Achieves dynamic strain rates from $\dot{\epsilon} \sim 10^3$ to 1.2×10^4 s⁻¹ under strictly verified 1D wave propagation with minimal Pochhammer-Chree dispersion.
- High-Frequency Diagnostics:** Instrumented with high-bandwidth semiconductor strain gauges at bar midpoints; isolates incident (σ_i), reflected (σ_r), and transmitted (σ_t) stress waves digitized at 10 MHz.
- High-Throughput Testing Protocol:** Modular alignment guides and precision collets enable rapid shot cycles (> 10 shots/hr), providing dense empirical datasets for statistical uncertainty bounds.

Abaqus/Explicit Digital Twin Modeling

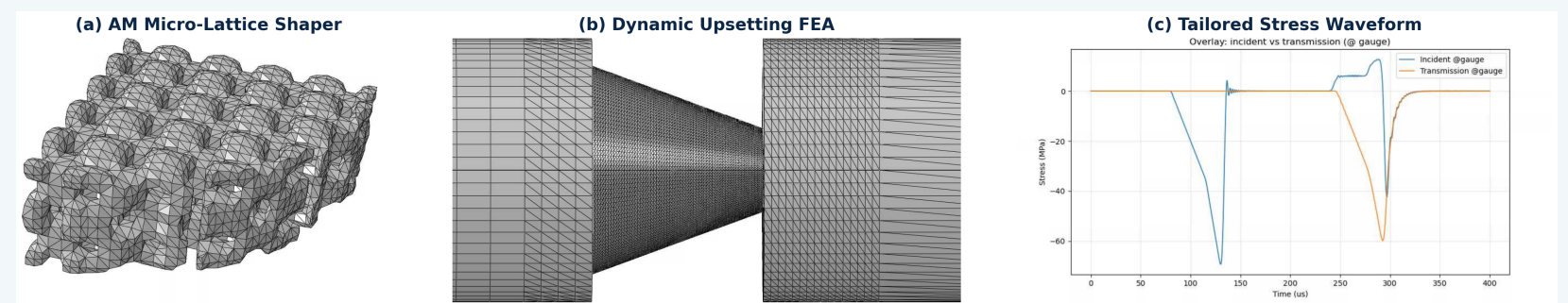
COMPUTATIONAL TWIN



- Multi-Symmetry FE Formulations:** 3D quarter- and full-symmetry models constructed in Abaqus/Explicit with C3D8R hex elements for dispersive wave propagation, striker impact, and platen boundary condition verification.
- Pre-Shot Waveform Optimization:** Virtual striker velocity and boundary impedance sweeps eliminate trial-and-error testing and maximize constant-strain-rate deformation duration.

Innovative Pulse Shaping for Tailored Response

TAILORED WAVEFORMS



- AM Micro-Architected Shapers:** Additively manufactured micro-lattices (nTopology/Abaqus) engineered to eliminate high-frequency Pochhammer-Chree wave dispersion and stress spikes.
- Constant Strain-Rate (CSR) Control:** Programmable rise times dynamically tailor the incident pulse to ensure force equilibrium and constant $\dot{\epsilon}$ deformation across soft and hard materials.

1. Virtual Pulse Shaping

Abaqus sweeps shaper geometry & velocity

2. Specimen & Shaper Prep

Wire-EDM precision fabrication & alignment

3. Dynamic Mini-SHPB Run

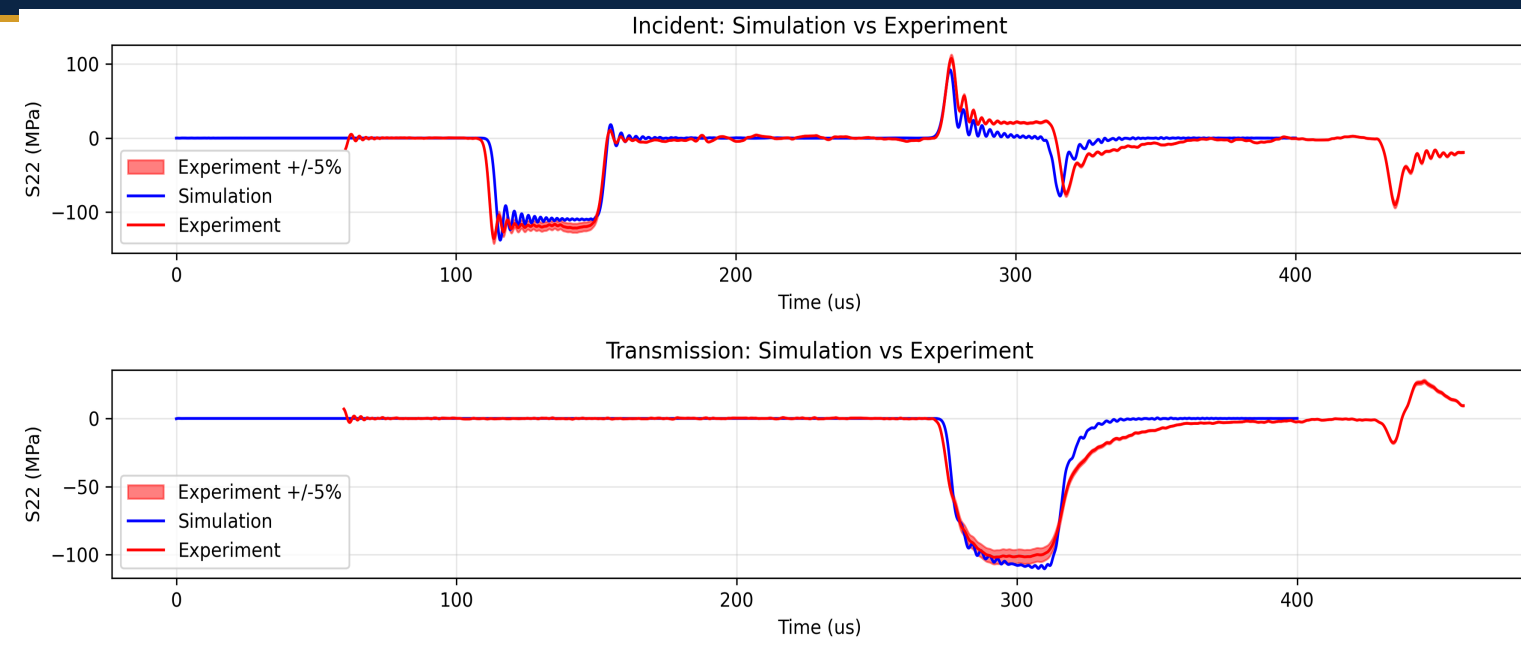
10 MHz strain acquisition + Nova S20 video

4. Model Calibration

Extract σ - ϵ curves & refine digital twin

Dynamic Wave Alignment & Validation

WAVE PROPAGATION



- Dispersion-Corrected Voltage Signals:** Incident and transmitted pulses captured at 10 MHz sampling frequency; wave profiles time-shifted to specimen contact interface.
- Digital Twin Predictive Accuracy:** Tight overlay between finite element simulation predictions and experimental bridge voltages across the primary pulse duration.
- Dynamic Force Equilibrium:** Verified across incident and transmission bar platens ($P_1(t) \approx P_2(t)$), ensuring homogeneous stress distribution throughout deformation.
- Model Refinement Feedback:** Waveform discrepancies directly guide the identification of rate-dependent strain hardening and thermal softening parameters.

Photron FASTCAM Nova S20 High-Speed Video

OPTICAL DIAGNOSTICS

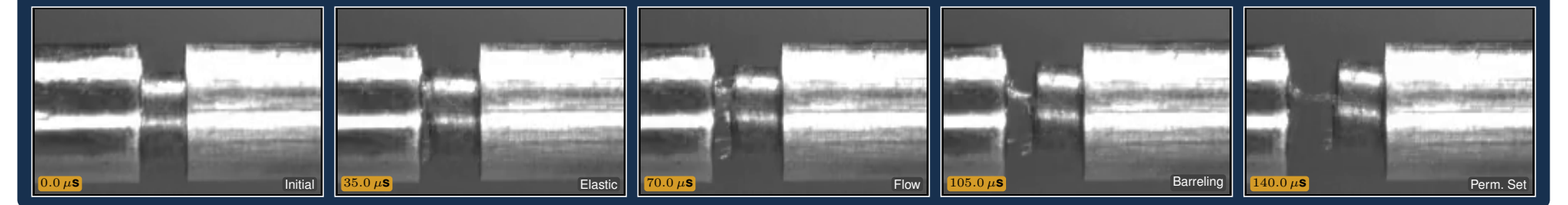


Optical Configuration & Synchronization:

The Photron FASTCAM Nova S20 is coupled with macro-telephoto optics and high-intensity LED matrix illumination, triggered synchronously with the incident bar strain bridge.

- Framing Rates:** 100,000 to 200,000 fps with sub- μ s exposure to freeze dynamic wave motion.
- Alignment Verification:** Real-time visualization guarantees coaxial platen impact without bar bending or eccentric loading.

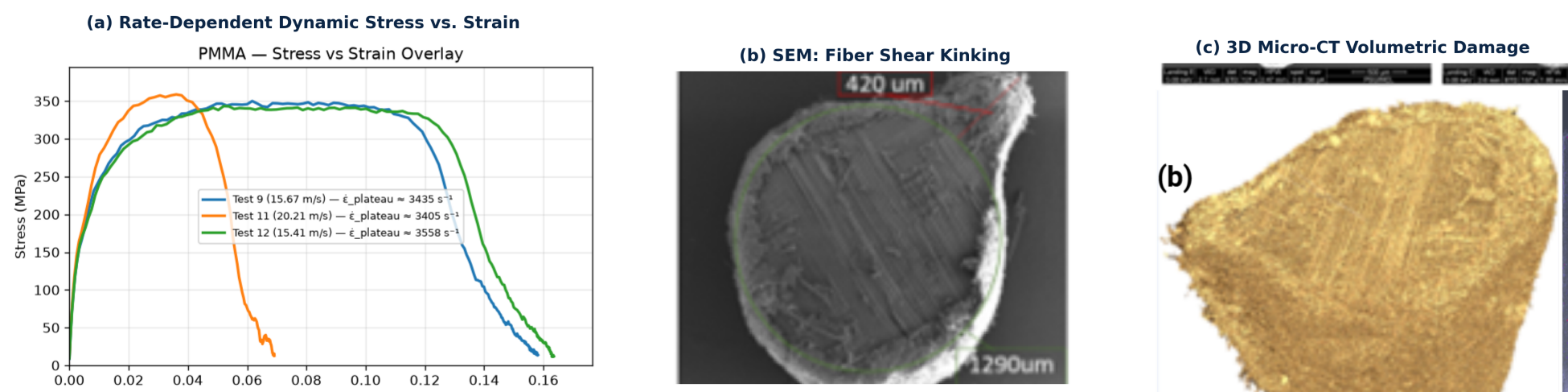
High-Speed Dynamic Compression Sequence ($\dot{\epsilon} \sim 4000$ s⁻¹ OFHC Copper on Ø3.17 mm C350 Bars)



- Deformation Kinematics:** In-situ high-speed sequence captures progressive axial upsetting, uniform radial expansion, and plastic barreling without shear localization.
- Dynamic Contact & Interface Uniformity:** Sub-microsecond exposure verifies planar platen seating and absence of specimen tilt or reverberating acoustic reflections during wave entry.
- Full-Field Optical Strain Tracking:** Synchronous frame recordings enable digital image correlation (DIC) to cross-validate bar 1D wave reduction against true surface kinematics.

Rate-Dependent Response & Failure Diagnostics

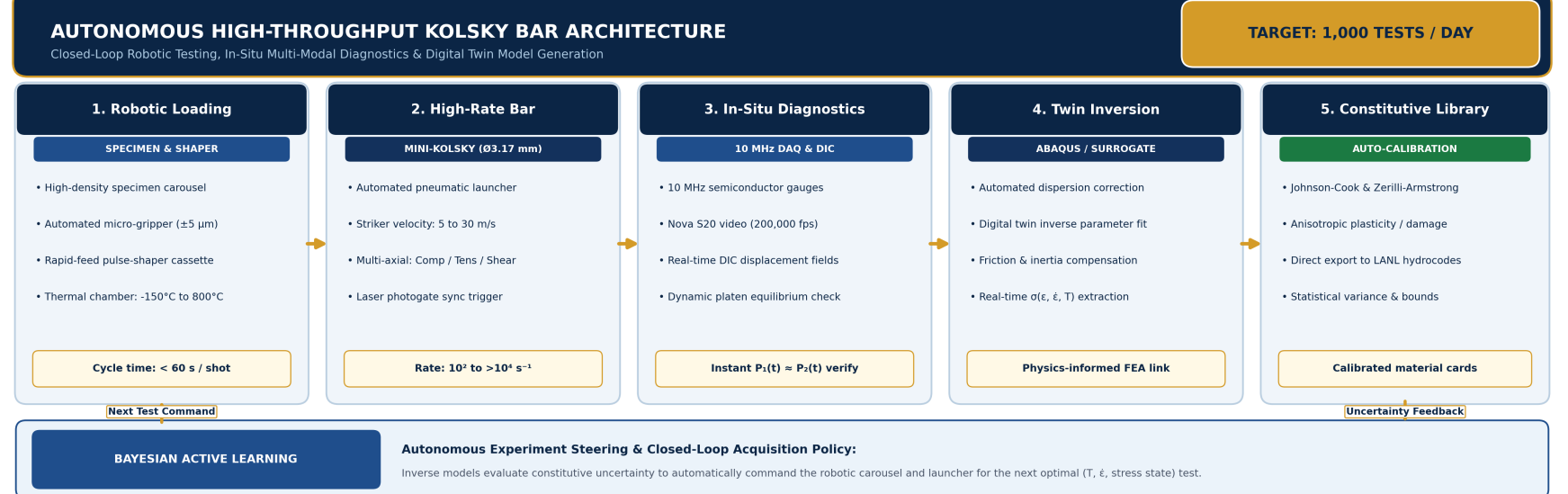
MATERIAL BEHAVIOR



- Extended Constant Strain-Rate Plateau:** Tailored pulse shaping yields a flat strain-rate plateau ($\dot{\epsilon} \approx 3400$ – 3560 s⁻¹), ensuring constitutive data is undistorted by dynamic effects.
- Dynamic Yield & Flow Enhancement:** PMMA specimens exhibit distinct rate hardening, elevating flow stress to ~ 345 MPa at high rates compared to quasi-static yield values (~ 100 MPa).
- Micro-Scale Failure Mechanisms:** Post-test SEM imaging reveals transverse fiber shear kinking, matrix crushing, and dynamic spallation at rates $> 10^3$ s⁻¹.
- Symbiotic Digital Twin Feedback:** Experimental flow stress curves and 3D micro-CT damage fields directly inform and calibrate anisotropic plasticity and dynamic failure criteria in LANL penetration simulations.

Future Horizons: Autonomous Kolsky Bar System

1,000 TESTS / DAY



- Autonomous 1,000-Tests/Day Throughput:** High-density robotic carousels, automated pick-and-place micro-grippers (± 5 μ m), and rapid-cycle pulse-shaper cassettes achieve < 60 s shot turnaround.
- Multi-Condition Parametric Testing:** Rapid automated sweeps across wide thermal bounds (-150°C to 800°C), strain rates (10^2 – 10^4 s⁻¹), and multi-axial stress states (compression, shear, tension).
- Active Learning Experiment Steering:** Bayesian acquisition algorithms identify high-uncertainty regions in constitutive space, automatically commanding the next test conditions to maximize information gain.
- Automated Rate-Dependent Model Discovery:** Synchronized 10 MHz DAQ and DIC feed into automated inverse FE solvers, auto-calibrating Johnson-Cook and damage models for direct LANL hydrocode deployment.